

Degenerate Bose gas and Bose-Einstein condensation

We know that $\bar{n}_{\vec{k}} = \frac{g}{e^{\beta(\epsilon_{\vec{k}} - \mu)} - 1}$, $g = \text{spin degeneracy} = 2S+1$

Since $\bar{n}_{\vec{k}}$ cannot be negative for any $\vec{k}$, $\mu \leq \min_{\vec{k}} \epsilon_{\vec{k}} = 0$ if $\epsilon_{\vec{k}} = \frac{\hbar^2 k^2}{2m}$
 $\Rightarrow z = e^{\beta\mu} \leq 1$, z is the fugacity.

Thus, for the ideal Bose gas, $0 \leq z \leq 1 \Leftrightarrow -\infty < \mu \leq 0$.

For a given value of μ (or z), the total number of particles is given by,

$$N = \sum_{\vec{k}} \bar{n}_{\vec{k}} = g \frac{V}{(2\pi)^3} \int d^3k \frac{1}{\frac{1}{z} e^{\beta \hbar^2 k^2 / 2m} - 1}$$
$$= \frac{gV}{(2\pi)^3} \int_0^\infty \frac{4\pi k^2 dk}{\frac{1}{z} e^{\beta \hbar^2 k^2 / 2m} - 1}$$

$$\text{Let } \frac{\beta \hbar^2 k^2}{2m} = x^2 \Rightarrow k = x \sqrt{\frac{2mk_B T}{\hbar^2}} = \frac{2\sqrt{\pi}}{\lambda_{th}} x,$$

where $\lambda_{th} = \sqrt{\frac{2\pi\hbar^2}{mk_B T}}$ = Thermal de Broglie wavelength.

$$\Rightarrow N = \frac{gV}{2\pi^2} \left(\frac{2\sqrt{\pi}}{\lambda_{th}} \right)^3 \int_0^\infty \frac{x^2 dx}{\frac{1}{z} e^{x^2} - 1}$$

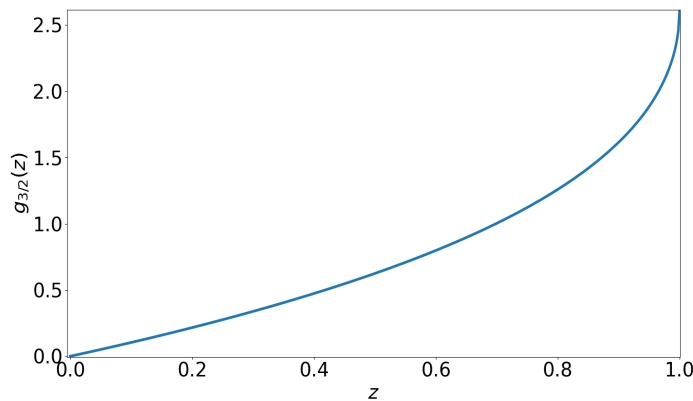
$$\Rightarrow \frac{n \lambda_{th}^3}{g} = \frac{4}{\sqrt{\pi}} \int_0^\infty \frac{x^2 dx}{\frac{1}{z} e^{x^2} - 1} = \text{function of } z \text{ only} \quad \left(n = \frac{N}{V} \right)$$

$$\text{Let } g_{3/2}(z) \equiv \frac{4}{\sqrt{\pi}} \int_0^\infty \frac{x^2 dx}{\frac{1}{z} e^{x^2} - 1} = \sum_{l=1}^\infty \frac{z^l}{l^{3/2}} \quad (\text{we will not use this})$$

$$\therefore \frac{n \lambda_{th}^3}{g} = g_{3/2}(z)$$

For a given temperature and density, we can calculate the L.H.S. of the above equation.

We can plot the R.H.S. as a function of z and the value of z for which $g_{3/2}(z)$ matches the value we evaluated using the LHS gives us the value of the fugacity or the chemical potential.



It is clear that

$$0 \leq g_{3/2}(z) \leq 2.612$$

$$g_{3/2}(1) = \zeta(3/2) = 2.612$$

↑
Riemann Zeta
function

What if $LHS > 2.612$?

This would be a chemical potential outside the physically allowed range (or a mode with negative occupancy).

In that situation, the value of the fugacity saturates to one and the chemical potential becomes zero and the particles occupy modes in accordance with the Bose-Einstein distribution formula and the excess particles occupy the lowest available energy state ($\vec{k}=0$).

This is known as the **Bose-Einstein condensation**.

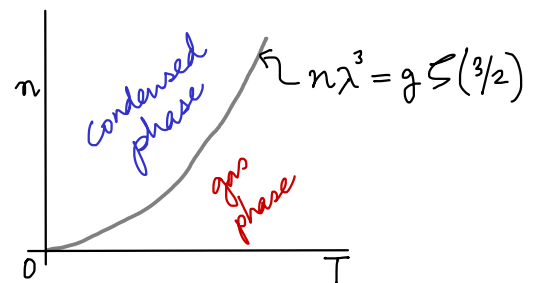
The lowest energy state gets macroscopically occupied.

The Bose-Einstein critical temperature

For a given density, the Bose-Einstein critical temperature can be obtained using,

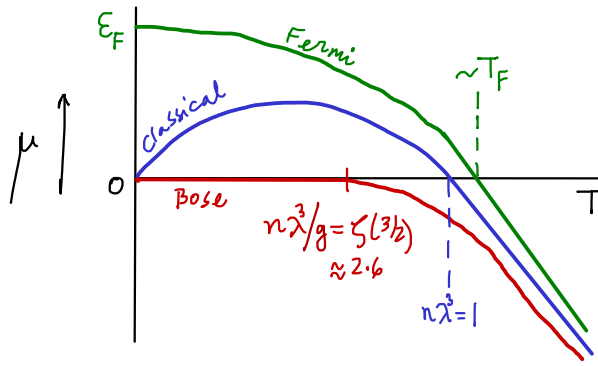
$$n \lambda_{th}^3(T_c) / g = \zeta(3/2) = 2.612$$

$$\Rightarrow \frac{n}{g} \left(\frac{2\pi\hbar^2}{mk_B T_c} \right)^{3/2} = \zeta(3/2)$$



Similarly, for a given temperature, the critical density n_c can be calculated using the same relation.

Comparison of chemical potential of ideal gases



The condensate

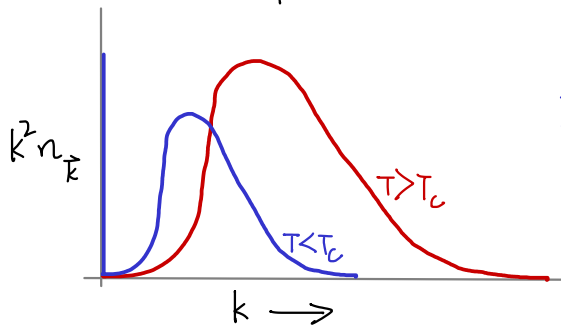
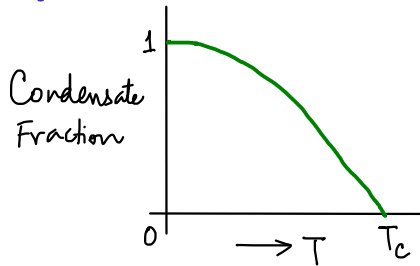
The number of particles that go into the condensate ($\vec{k}=0$) state is

$$n_0 = \left[n - g \zeta(3/2) / \lambda^3 \right] V$$

$$\Rightarrow \text{Fraction of particles in the condensate} = \frac{n_0}{N} = 1 - \frac{g \zeta(3/2)}{n \lambda^3}$$

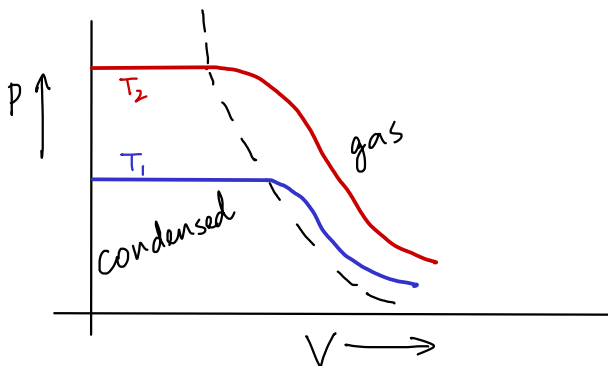
$$= 1 - \frac{g \zeta(3/2)}{n \lambda_c^3} \left(\frac{\lambda_c^3}{\lambda} \right)$$

$$= 1 - \left(\frac{T}{T_c} \right)^{3/2}$$



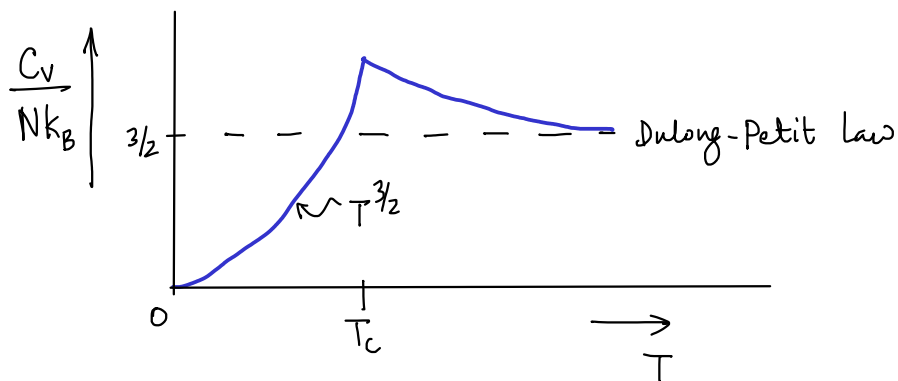
Below T_c a δ -function appears at $\vec{k}=0$.

Other things



Volume at the transition line $= V^*$

$$\frac{N}{V^*} \lambda^3 = \zeta(3/2) g$$

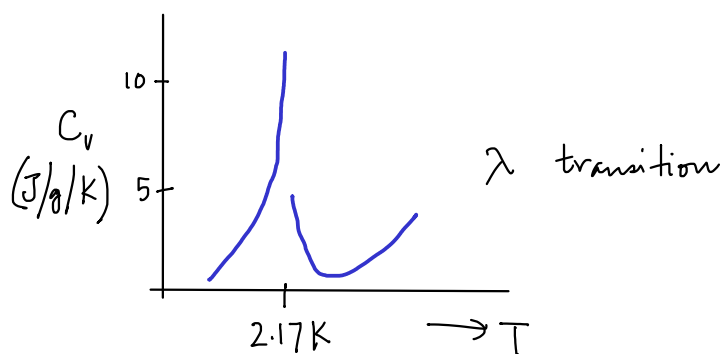


Liquid Helium

The ${}^4\text{He}$ atom is a boson. It is therefore expected to undergo Bose-Einstein condensation at low temperatures. Because of interactions, liquid ${}^4\text{He}$ cannot be treated as an ideal Bose gas.

Below T_c , $C_v \propto T^3$

${}^4\text{He}$ attains a superfluid state below T_c
 $\downarrow$
 vanishing viscosity



$$T_c = 2.17\text{ K}, \quad n_c = 2.16 \times 10^{22} / \text{cm}^3$$

Using the ideal Bose gas expression,

$$T_c = \left[\left(\frac{1}{g \zeta(3/2)} \right)^{2/3} \frac{2\pi\hbar^2}{mk_B} \right] n^{2/3} \approx 3.1\text{ K}$$

(Here m is the mass of a ${}^4\text{He}$ atom and $g=1$.)